



Characterization of optical diffraction by single nanochannel for aL–fL sample detection in nanofluidics

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Abstract

Nanofluidics which integrates analytical systems in 10^1 – 10^3 nm space provides ultra-sensitive analyses at a single-cell and single-molecule level. One of the key technologies for nanofluidics is the ultra-sensitive detection method; however, the ultra-small volume at aL–fL scale makes it challenging. Recently, we have developed a non-fluorescent molecule detection method for nanofluidics called photo-thermal optical diffraction (POD) which utilizes the photo-thermal effect of target molecules and optical diffraction by a single nanochannel. To improve the performance of such diffraction-based detection methods, the design and optimization of optical diffraction are essential. However, it is unknown whether the optical diffraction by a single nanochannel follows general diffraction theory because liquid properties change in the ultra-small space. In this study, we elucidated optical diffraction by a single nanochannel from theoretical calculations and experiments. Our experiments revealed the effect of channel size, channel position, and solvents in the nanochannel, which showed good agreement with proposed theoretical calculations. We also revealed no or little change of refractive index of water in the nanochannel compared with that in the bulk. Finally, we confirmed that the POD signal was proportional to the diffracted light intensity, and the calculated limit of detection of POD was 7.0×10^{-7} RIU in a detection volume of 0.23 fL. Our theoretical calculations and experimental results can be widely applied to the design and optimization of detection methods using optical diffraction by nanochannels and nanostructures.

Keywords Nanofluidics · Nanochannel · Diffraction · Photo-thermal effect · Refractometry

1 Introduction

There has been rapid progress in the field of microfluidics which integrates micrometer-scale channels on a centimeter-scale substrate for chemical and biological processing. Recent advances in fabrication technologies provide further miniaturization of the analytic spaces to 10^1 – 10^3 nm scale (nanofluidics) (Napoli et al. 2010; Mawatari et al. 2014). The characteristics of nanofluidics are large surface-to-volume ratio, ultra-small sample volume (pL–aL), and comparable size with vesicles, DNAs, and protein molecules. Several unique applications utilizing these characteristics have been developed. For separation science, nano-chromatography and molecular sieving by nanostructures have been

developed (Huh et al. 2007; Shimizu et al. 2019). For bio-molecule analyses and diagnoses, single-molecule immunoassay, single-cell analysis, and DNA sequence by nanopore have been developed (Huang et al. 2010; Shirai et al. 2018; Nakao et al. 2019). For energy conversion, energy harvesting by nanopore and micro fuel cell device have been developed (Guo et al. 2010; Pihosh et al. 2017). For fundamental science, unique properties in nanochannels have been reported: fast enzymatic reaction, ionic mobility and distribution, solute and solvent transport through nanochannels, and structural change of water (Schoch et al. 2008; Chinen et al. 2012; Wang et al. 2013; Mawatari et al. 2018). As described above, nanofluidics plays a significant role in fundamental sciences and applications.

Detection or sensing methods are essential for analyzes. However, it is challenging for nanofluidics because sample volume and the number of molecules are quite small compared with those for bulk-scale measurements. For example, the volume of a nanochannel with 100 μ m length and 100 nm width and depth is 1 fL. If 1 nM solution is

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introduced to the channel, the number of the molecules in the channel is less than single-molecule (0.6 molecules); nanofluidics requires highly sensitive detection or sensing methods. Laser-induced fluorescence is commonly used as a sensitive molecule detection method (Brooks Shera et al. 1990), while it requires tedious labeling processes. Photo-thermal spectroscopy (PTS) is widely used as a highly sensitive and label-free detection method. PTS measures local temperature change or refractive index change caused by the light absorption and heat generation by target samples (photo-thermal effect). Several photo-thermal detection methods have been developed such as thermal lens detection, photo-acoustic detection, and photo-thermal interferometry (Hoelen et al. 1998; Tokeshi et al. 2001; Shimizu et al. 2009). However, applying these techniques to nanofluidics is difficult due to the thermal loss by fast thermal diffusion from the channels to the substrates (e.g., glass) (Shimizu et al. 2017).

Recently, we have developed a non-fluorescent molecule detection method for nanofluidics called photo-thermal optical diffraction (POD) which utilizes the photo-thermal effect of target molecules and optical diffraction by a single nanochannel (Tsuyama and Mawatari 2019). By utilizing POD, non-fluorescent molecule detection in $400 \text{ nm} \times 400 \text{ nm}$ channel has been achieved with the LOD of 500 molecules. POD measures diffracted light intensity change caused by the difference of physical properties of the solvent in the channel and those of the glass substrate. To clarify the principle of POD and improve the detection performance, it is important to understand the optical diffraction by a single nanochannel. Although optical diffraction by periodic structures such as diffraction grating has been theoretically and experimentally investigated (Tufekcieva and Jonoska 1997; Beleggia 2008), there are few reports on optical diffraction by non-periodic structures. Furthermore, as mentioned above, unique liquid properties (dielectric constant, viscosity, etc.) in nanochannels have been reported; it is unknown whether the optical diffraction by a single nanochannel follows general diffraction theory and liquid properties in the bulk.

In this study, we clarified the characteristics of optical diffraction by a single nanochannel from theoretical calculations and experiments. First, the theoretical formula of optical diffraction by nanochannels was derived based on the Fresnel–Kirchhoff diffraction theory. Then, we confirmed the effect of channel size, laser condition, and solvents on diffracted light intensity distribution by theoretical calculations and experiments. Furthermore, we investigated the electronic polarization of water in the nanochannel from the diffracted light intensity. Finally, we confirmed the relationship between diffracted light intensity and the POD signal.

2 Theory

To understand the optical diffraction by a single nanochannel, it is necessary to calculate the intensity distribution of the laser beam after passing through the nanochannel. We derived a theoretical formula based on the Fresnel–Kirchhoff diffraction theory (Max and Emil 1999). The schematic diagram of a single nanochannel diffraction measurement is shown in Fig. 1. The electric field of focused TEM₀₀ mode laser beam at the channel position can be expressed as:

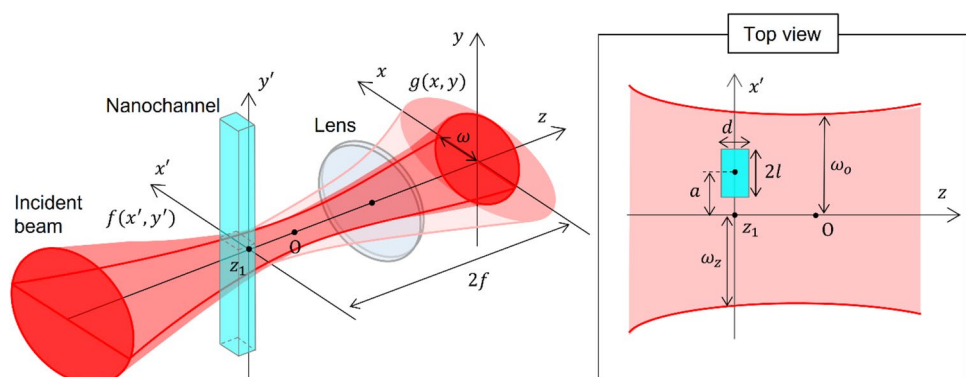
$$f(x', y') = \sqrt{\frac{2P}{\pi\omega_z^2}} \exp \left[-i \left\{ kz_1 + \arctan \left(\frac{z_1}{z_0} \right) - \frac{k(x'^2 + y'^2)}{2R(z_1)} \right\} - \frac{x'^2 + y'^2}{\omega_z^2} \right] \quad (1)$$

where P is the total power of the laser beam, $k = 2\pi/\lambda$ is the wavenumber, $R(z_1)$ is the phase front radius, z_1 is the distance from the beam waist to the channel position, $z_0 = k\omega_0^2/2$ is the confocal parameter, ω_0 is the radius ($1/e^2$) of the laser beam at its waist, and ω_z is the radius of the laser beam at the channel position defined by:

$$\omega_z = \omega_0 \sqrt{1 + \left(\frac{z_1}{z_0} \right)^2} \quad (2)$$

The depth of a nanochannel is equivalent or smaller than the wavelength of visible light, and it is one order of magnitude smaller than the confocal length of the laser beam.

Fig. 1 The schematic diagram of a single nanochannel diffraction



Therefore, we assumed that the nanochannel works as a “phase filter”, changing the phase of the beam by passing through the channel. Under this assumption, the phase filter function $t(x', y')$ representing the relative phase shift by the nanochannel with width $2l$ and depth d is expressed as follows:

$$t(x', y') = \begin{cases} e^{i\theta} & (a - l \leq x' \leq a + l) \\ 1 & (x' < a - l, x' > a + l) \end{cases} \quad (3)$$

where

$$\theta = \frac{2\pi}{\lambda} \cdot (n_s - n_g) \cdot d \quad (4)$$

a is the distance from the center of the beam to the channel position, n_s is the refractive index of the solvent in the channel, and n_g is the refractive index of the glass. The electric field after passing through the channel is given by the product of incident electric field $f(x', y')$ and phase filter function $t(x', y')$.

From the Fresnel–Kirchhoff diffraction theory, the electric field of the laser beam at a back focal plane of a lens is given by Fourier transfer of the electric field at a front focal plane of the lens. If the front focal plane of the lens is set at the channel position, the electric field at the back focal plane, $g(x, y)$ is given by:

$$g(x, y) = \frac{1}{i\lambda f} \iint f\left(\frac{x'}{\lambda f}, \frac{y'}{\lambda f}\right) t\left(\frac{x'}{\lambda f}, \frac{y'}{\lambda f}\right) \exp\{-i2\pi(x'x + y'y)\} dx' dy' \quad (5)$$

where f is the focal length of the lens. In this calculation, the lens diameter is assumed to be sufficiently large. By using Eqs. (1), (3), and (5), we get the following expression for the electric field of the laser beam at the back focal plane of the lens:

$$g(x, y) = \sqrt{\frac{2P}{\pi\omega^2}} \exp\left(-\frac{x^2+y^2}{\omega^2}\right) \{1 + (e^{i\theta} - 1)u(x, l)\} \quad (6)$$

where

$$u(x, l) = \frac{1}{2} \left\{ \operatorname{erf}\left(\frac{a+l}{\omega_0 z_r} + i\frac{xz_r}{\omega}\right) - \operatorname{erf}\left(\frac{a-l}{\omega_0 z_r} + i\frac{xz_r}{\omega}\right) \right\} \quad (7)$$

$\operatorname{erf}(x)$ is error function, ω is the radius of the laser beam at the focal plane, z_r is defined by

$$z_r = \sqrt{1 - i\frac{z_1}{z_0}} \quad (8)$$

The intensity distribution of the laser beam $I(x, y)$ is given by the square of the electric field $g(x, y)$, which is expressed as:

$$I(x, y) = \frac{2P}{\pi\omega^2} \exp\left\{-\frac{2(x^2+y^2)}{\omega^2}\right\} \times [1 + 2|u(x, l)|^2(1 - \cos\theta) - 2\operatorname{Re}[u(x, l)](1 - \cos\theta) - 2\operatorname{Im}[u(x, l)]\sin\theta] \quad (9)$$

The first part of Eq. (9) describes the transmitted light, intensity distribution of laser beam without nanochannel. The second part describes the diffracted light. The third and fourth parts describe interference between transmitted light and diffracted light. This theoretical calculation can be applied to other nanostructures such as diffraction grating by changing $t(x', y')$, which leads to the change of $u(x, l)$. For example, the nanochannel diffraction grating with n channels which width is l and gap is $2l$ gives following electric field at a back focal plane when TEM₀₀ mode laser beam is focused at the symmetry center of the grating:

$$g(x, y) = \sqrt{\frac{2P}{\pi\omega^2}} \exp\left(-\frac{x^2+y^2}{\omega^2}\right) \{1 + (e^{i\theta} - 1) \sum_{m=1}^n u_m(x, l) (-1)^{m+n}\}, \quad (10)$$

where

$$u_m(x, l) = \frac{1}{2} \left\{ \operatorname{erf}\left(\frac{(2m-1)l}{\omega_0 z_r} + i\frac{xz_r}{\omega}\right) - \operatorname{erf}\left(-\frac{(2m-1)l}{\omega_0 z_r} + i\frac{xz_r}{\omega}\right) \right\}. \quad (11)$$

If the optical diffraction by a single nanochannel is used as a sensor or detector, the signal is diffracted light intensity change (ΔI) caused by the refractive index change in the channel (Δn_s) and the glass substrate (Δn_g). Assuming that Δn_s and Δn_g are small and $z_1 = 0$, $\Delta I(x, y)$ is expressed as follows:

$$\Delta I(x, y) \approx \frac{dI(x, y)}{d\theta} \cdot \Delta\theta = \frac{4Pd(\Delta n_s - \Delta n_g)}{\lambda\omega^2} \exp\left\{-\frac{2(x^2+y^2)}{\omega^2}\right\} \cdot 2u(x, l)\{u(x, l) - 1\}\sin\theta. \quad (12)$$

The detection performance can be estimated from the ratio of signal (ΔI) and background (I). Assuming that the effect of transmitted light is negligible, the S/B ratio at a diffracted light region is given by:

$$\frac{S}{B} = \frac{\sin\theta}{1 - \cos\theta} \cdot \frac{2\pi d(\Delta n_s - \Delta n_g)}{\lambda} \approx \frac{2(\Delta n_s - \Delta n_g)}{(n_s - n_g)} (\theta \ll 1). \quad (13)$$

These theoretical formulas indicate several important features of nanochannel diffraction. First, the diffraction pattern does not depend on the refractive indices if the channel depth is shallow. The relative value of the spot radius and the channel width determines the diffraction pattern. Second, diffracted light intensity depends on the refractive indices and channel depth. Third, the S/B ratio does not largely depend on the channel width and depth, which is also reported by other groups (Ajiri et al. 2017; Tu et al. 2019). Note that we need to consider the spatial distribution of Δn_s and Δn_g for accurately calculating the signal of POD.

3 Experimental

3.1 Experimental setup

The experimental setup and optical systems for diffracted light intensity measurement are shown in Fig. 2a, b. A He–Ne laser (633 nm, 25-LHP-925, Melles Griot Inc., USA) was introduced to a microscope (BX51, Olympus Corp., Japan) and focused on a nanochannel by an objective lens (20 $\times$, NA=0.45). A light chopper (5584A, NF Corp., Japan) modulated the intensity of the beam at a modulation frequency of 1.1 kHz for synchronized detection. The intensity of the beam was 1–5 mW under the objective lens. A nanofluidic device was fixed on a 2-D stage (BIOS-235S,

SIGMA KOKI CO., LTD., Japan) which has 100 nm steps for x- and y-direction to precisely adjust the channel position. After passing through the channel, a large NA objective lens (NA=0.90) collected transmitted and diffracted beams. A slit cut the transmitted beam, and an interference filter cut the external light. The remaining diffracted beam was collected on a photodiode (ET-2030, Electro-Optics Technology Inc., USA) and fed into a lock-in amplifier (LI5660, NF Corp., Japan). The diffracted beam intensity was measured by extracting the component synchronized to the chopper frequency. The time constant of the lock-in amplifier was set to 1 s. A data logger recorded the signal intensity every 10 ms. Figure 2c, d show typical images of the beam spot after passing through the collection lens. The channel was empty (filled with air), and channel width and depth were 400 nm. After passing through the nanochannel, the diffracted light was observed in the direction perpendicular to the longitudinal direction of the channel as predicted by theory.

3.2 Device fabrication and sample injection

The detailed fabrication method was explained previously (Tsukahara et al. 2009). Briefly, nanochannels were fabricated on a fused-silica glass substrate (70 $\times$ 30 mm², VIOSIL-SQ, Shin-Etsu Chemical Co., Ltd., Japan) by electron-beam lithography (ELS-7500, Elionix Inc., Japan) and plasma etching (NE-550, ULVAC Inc., Japan). The width and depth of fabricated nanochannels were measured by scanning electron microscope (equipped with ELS-7500) and atomic force microscope (SPA-400, Seiko Instruments Inc., Japan), respectively. For sample injection to the nanochannels, U-shaped microchannels were fabricated on another glass substrate by photolithography and plasma etching. Inlet holes were drilled at the endpoint of each microchannel. The width and depth of microchannels were 500 μ m and 5 μ m. By thermal bonding of the two glass substrates, microchannels and nanochannels were connected. As shown in Fig. 2a, sample solutions were injected into nanochannels by a pressure-driven flow control system. Sample solutions were stocked in a vial connected to a pressure controller (MFCS-EZ, Fluigent Inc.) and the inlet of microchannels. By adjusting and applying the pressure to the vial, the flow rate in the nanochannels was precisely controlled.

3.3 Sample preparation

To clarify the effect of refractive index for the optical diffraction by a single nanochannel, we used seven different pure solvents with different refractive indices: water ($n=1.333$), ethanol ($n=1.360$), butyl acetate ($n=1.390$), 1-hexanol ($n=1.415$), cyclohexanone ($n=1.448$), DMSO ($n=1.476$), and toluene ($n=1.493$). Refractive indices of

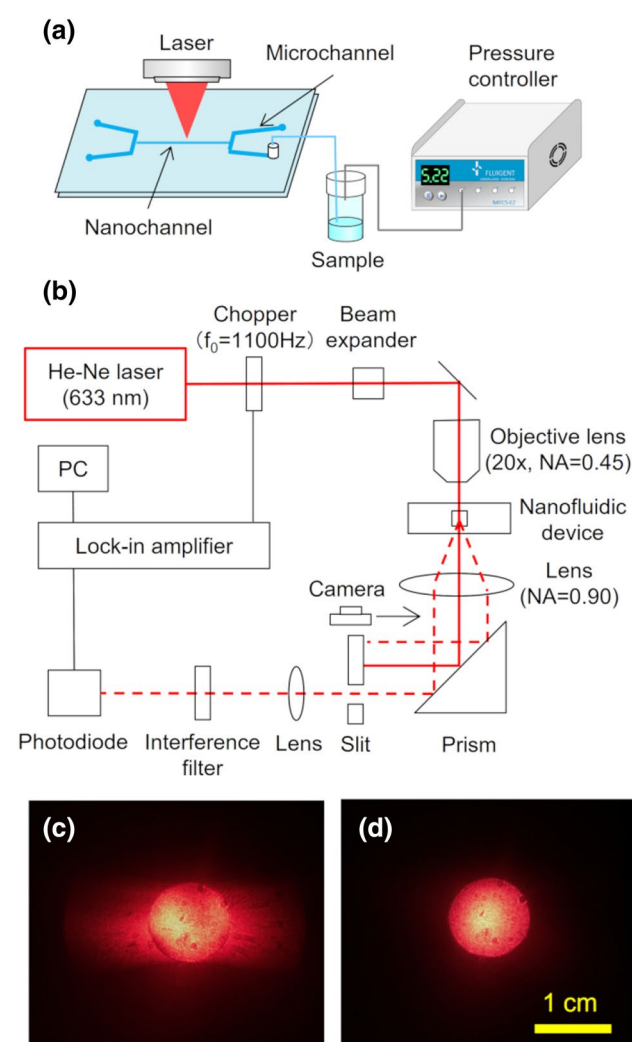


Fig. 2 **a** Experimental setup and **b** optical systems of nanochannel diffraction detection. Dashed lines indicate diffracted light by a nanochannel. **c** Typical image of optical diffraction by a single nanochannel with 400-nm width and depth. **d** Typical image of beam spot without channel

solvents were measured by refractometer (RA-130, Kyoto Electronics Manufacturing CO., LTD., Japan). Water was used as a solvent unless otherwise mentioned. For POD measurements, an aqueous solution of the non-fluorescent dye, Sunset Yellow FCF was introduced to nanochannels. All reagents were purchased from FUJIFILM Wako Pure Chemical Corporation, Japan.

4 Results and discussion

4.1 Evaluation of the channel-size effect

First, we investigated the relationship between channel width and detected diffracted light intensity. The measured range of channel width was 220–2100 nm, and the channel depth was fixed at 220 nm. These channels were filled with water. For theoretical calculations, we used the following numerical values. $\omega_0 = 1.1 \mu\text{m}$, $\lambda = 633 \text{ nm}$, $n_s = 1.33$, $n_g = 1.46$, $\omega = 2.5 \text{ mm}$, and the detection region was 5–15 mm from the center of the laser spot. To compare the theoretical calculations and experimental results, calculated and experimental values were normalized by their maximized value. Theoretical calculation and experimental results are shown in Fig. 3. Theoretical values were calculated by a custom-made VBA macro in Excel (Microsoft). As predicted from theoretical calculations, the diffracted light intensity was maximized at certain channel width. This tendency is explained as follows. For the narrower channel, most of the probe beam passes through the substrate, and the total phase difference decreases which is caused by the refractive index difference between water and glass. For wider channels, most of the probe beam passes through the channel, and the total phase difference also decreases. Thus, the optimum channel width depends on the spot size. For our current experimental

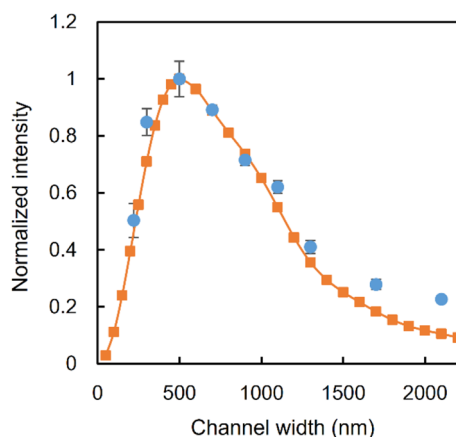


Fig. 3 Normalized diffracted light intensity versus channel width. Orange line represents theoretical calculations and blue dots represent experimental results

conditions, the optimum channel width was around 500 nm, which corresponds to $0.5 \omega_0$. Note that the optimum channel width also depends on other factors such as wavelength or detection region of the diffraction pattern. The reason for the difference between theoretical calculations and experimental results for wider nanochannels is under investigation. Also, calculated diffraction patterns for several channel widths are shown in Fig. 4. Calculated diffraction patterns show that the diffraction region becomes broader as channel width becomes smaller. This tendency is in good agreement with our previous experimental results (Tsuyama and Mawatari 2019).

Next, we investigated the relationship between channel depth and detected light intensity. The measured range of channel depth was 60–400 nm, and the channel width was fixed at 400 nm. These channels were also filled with water. As shown in Fig. 5, diffracted light intensity decreased proportionally to the square of the channel depth. This tendency is explained by the decrease of the phase difference between the probe beam passing through the channel and the substrate. These results show good agreement with theoretical calculations, suggesting our phase filter approximation was valid in the range of measured channel depth. The agreement of theory and experiment shows that theoretical calculation based on the Fresnel–Kirchhoff diffraction theory enables the design and optimization of the channel size. Note that we also need to consider the difference of thermal distribution by nanochannel sizes for POD experiments.

4.2 Evaluation of the channel-position effect

The size of nanochannels is equivalent to the size of the focused laser beam spot, and diffracted light intensity is

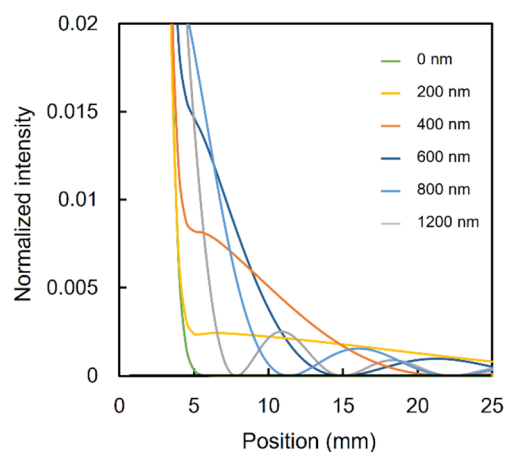


Fig. 4 Calculated diffraction patterns by a single nanochannel with different channel widths. The diffracted light intensity was normalized by the intensity at the center of the probe beam spot without channel

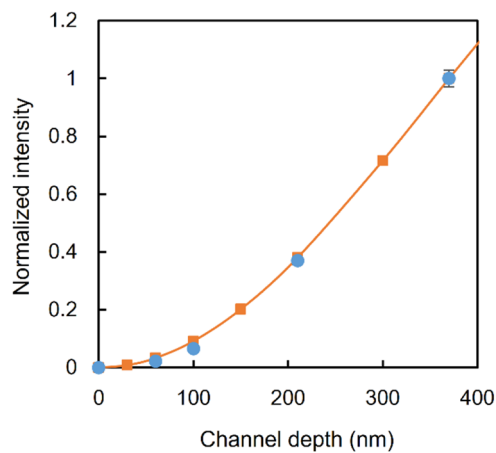


Fig. 5 Normalized diffracted light intensity versus channel depth. Orange line represents theoretical calculations and blue dots represent experimental results

sensitive to the position of the spot. Figure 6 shows the dependency of diffracted light intensity on the relative position of laser focus in the x axis direction, perpendicular to the longitudinal direction of the channel. We used a single nanochannel with 400-nm width and depth filled with water. The channel position was adjusted by the electric stage with 100-nm steps. Theoretical calculation and experimental results show that diffracted light intensity was maximized when the channel position was fixed at the position of laser focus, and it decreased as the distance between the channel and the laser focus (Δx) becomes longer. The full width at half maximum was approximately 1.4 μm .

Figure 7 shows the dependency of diffracted light intensity on the relative position of laser focus in the z axis

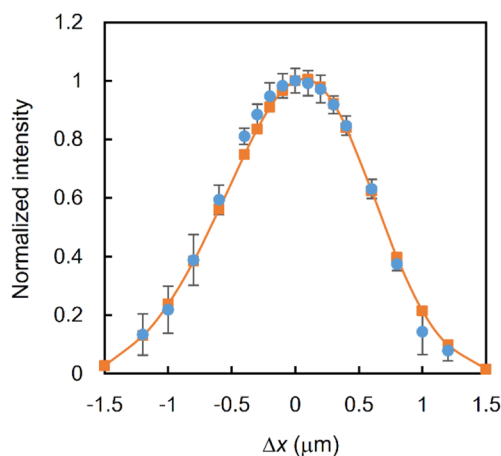


Fig. 6 Normalized diffracted light intensity versus relative position of laser focus in the x axis direction (Δx). Orange line represents theoretical calculations and blue dots represent experimental results

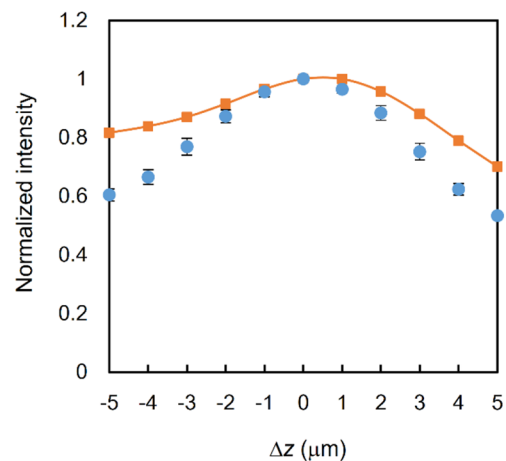


Fig. 7 Normalized diffracted light intensity versus relative position of laser focus in the z axis direction (Δz). Orange line represents theoretical calculations and blue dots represent experimental results

direction, parallel to the laser propagation direction. The channel position was adjusted with 1 μm steps. The diffracted light intensity was maximized at $\Delta z = 0$ and gradually decreased as Δz becomes larger. The full width at half maximum was approximately 10 μm . Compared with theoretical calculations, diffracted light intensity largely decreased for Δz . This might be derived from the overestimation of the spot size. The measured spot size is larger than the theoretical size due to the aberration. Therefore, the actual spot size is more sensitive to Δz than the theoretical one; diffracted light intensity is lower than the theoretical value at a position away from the laser focus.

These results show that the optimum position of laser focus is at the center of the channel, and precise position alignment is essential for single nanochannel diffraction. Larger spot size by lower NA objective lens increases the tolerance for the position, although it also affects diffracted light intensity.

4.3 Evaluation of the solvent effect

Diffracted light intensity largely depends on the refractive index of the solvent in the nanochannel. Figure 8 shows theoretical calculations and experimental results for seven different pure solvents with different refractive indices (water, ethanol, butyl acetate, 1-hexanol, cyclohexanone, DMSO, and toluene). Each solvent was introduced to the channel with 400 nm width and depth. The channel position was fixed at the position of the laser focus. For precise comparison, calculated values were fitted by the least-squares method. As predicted by the theory, diffracted light intensity changed with a refractive index, and the value was minimized when the refractive index of the solvent was close to that of the glass substrate ($n_g = 1.46$). These results also

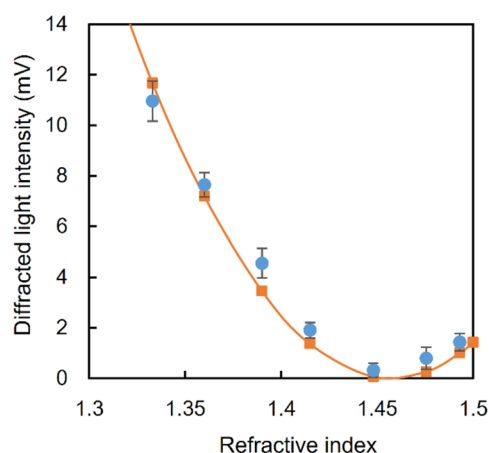


Fig. 8 Diffracted light intensity for refractive indices of solvents. Orange line represents theoretical calculations and blue dots represent experimental results

suggest that optical diffraction by a single nanochannel can be used for refractive index measurement of the solvent in the channel.

In our experiments, little difference between theory and experiment was observed for the nanochannel filled with water; the refractive index of water in the nanochannel is almost the same as that in the bulk, and the optical diffraction by a single nanochannel follows general diffraction theory. Although a decrease in dielectric constant (proportional to the square of the refractive index) has been reported for 10^2 nm channels (Hibara et al. 2002; Morikawa et al. 2015), they measured dielectric constant for low-frequency region where orientation polarization is dominant. Therefore, their results suggest that water confined in nanochannels has a highly ordered structure that consists of loosely coupled water molecules. In contrast, we measured the refractive index for the THz frequency region where electronic polarization is dominant. Thus, our results suggest that the electric polarization of water in the nanochannel is almost the same as that in the bulk.

4.4 POD signal dependency on the diffracted light

Finally, we checked the relationship between the POD signal and diffracted light intensity. The principle and experimental setup of POD was previously reported (Tsuyama and Mawatari 2019). As a sample solution, an aqueous solution of Sunset Yellow FCF was introduced to the nanochannel with 400 nm width and depth. Diffracted light intensity was adjusted by changing laser intensity and slit width. Experimental results are shown in Fig. 9. POD signal increased linearly with the diffracted light intensity, indicating that the design and optimization of

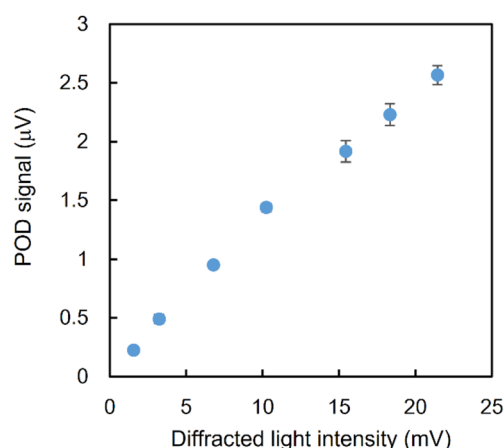


Fig. 9 POD signal for diffracted light intensity. Each value was measured by a lock-in amplifier at 1.1 kHz frequency modulation

optical diffraction are essential for realizing high detection performance. The ratio of signal to background (S/B) was 1.2×10^{-4} for 100 μM Sunset Yellow solution, and the calculated limit of detection of POD was 7.0×10^{-7} RIU (5 μM). The detection volume was 0.23 fL, determined from excitation beam spot diameter (1.4 μm) and channel width and depth (400 nm). These results show that the optical diffraction by a single nanochannel enables the measurement of ultra-small refractive index change by the photo-thermal effect of target molecules.

5 Concluding remarks

In this study, we theoretically and experimentally clarified the optical diffraction by a single nanochannel. Theoretical calculations showed good agreement with experimental results and can be used for the estimation of the optical diffraction by nanochannels with various material, shape, and size. It will be also applied to other nanostructures such as diffraction gratings or pillars. Our experimental results revealed no or little change of refractive index of water in the nanochannel compared with that in the bulk; the optical diffraction by nanochannels can be designed based on Fresnel–Kirchhoff diffraction theory and liquid properties measured in the bulk. The design and optimization of optical diffraction by a single nanochannel enables sensitive molecule detection and refractive index measurement for aL–fL detection volume. As mentioned in the theory section, the S/B ratio of ΔI measurement does not depend on the channel width and depth, although diffracted light intensity decreases for smaller channels. Thus, ΔI measurement for smaller channels can be realized with a similar performance by using a high power laser or optimized optical setup. Precise

position control of the laser focus and stabilization of laser intensity distribution enables further sensitivity improvement.

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